ECO208 Macroeconomic Theory

Lecture 20

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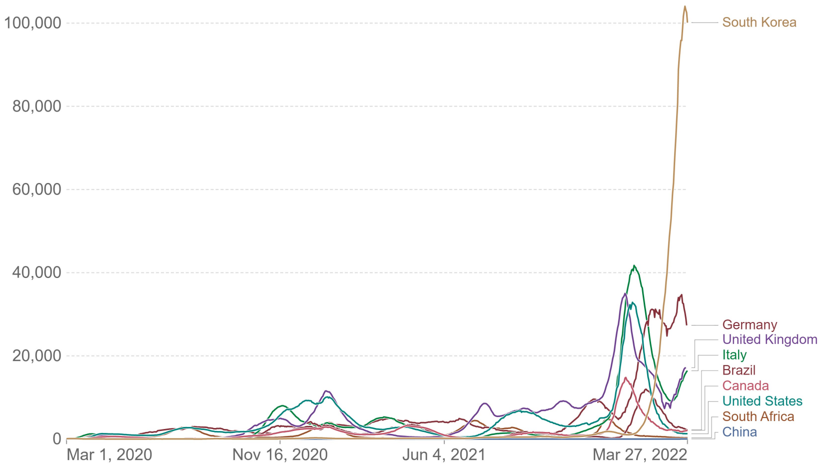
UTM, March 29, 2022

The COVID-19

Global Pandemic and Macroeconomic Policy
Part I

Biweekly confirmed COVID-19 cases per million people

Biweekly confirmed cases refers to the cumulative number of cases over the previous two weeks.

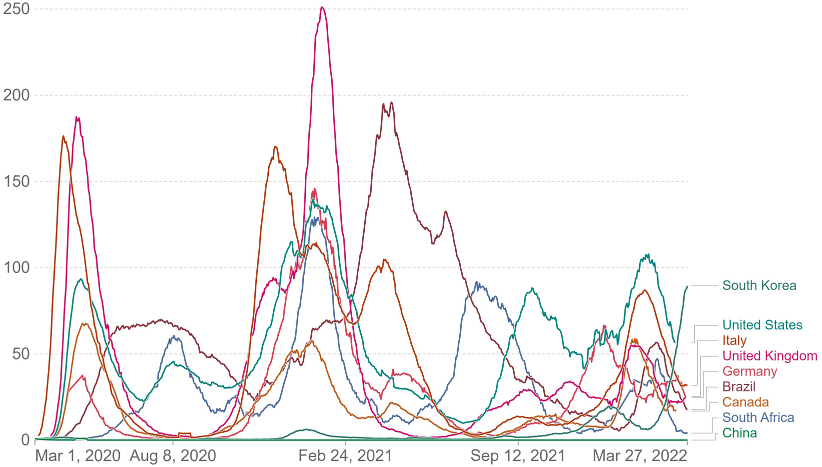


Source: Johns Hopkins University CSSE COVID-19 Data – Last updated 28 March, 19:05 (London time)

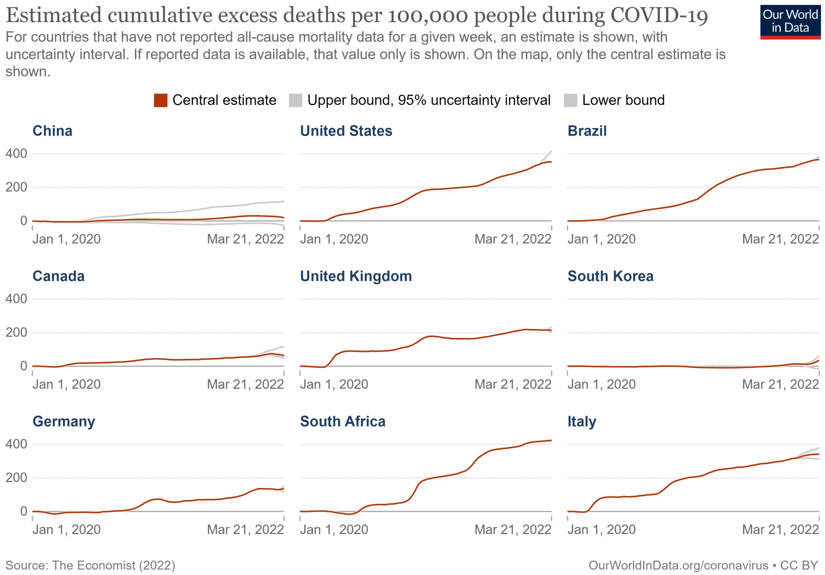
Biweekly confirmed COVID-19 deaths per million people

Biweekly confirmed deaths refer to the cumulative number of confirmed deaths over the previous two weeks.

Our World in Data

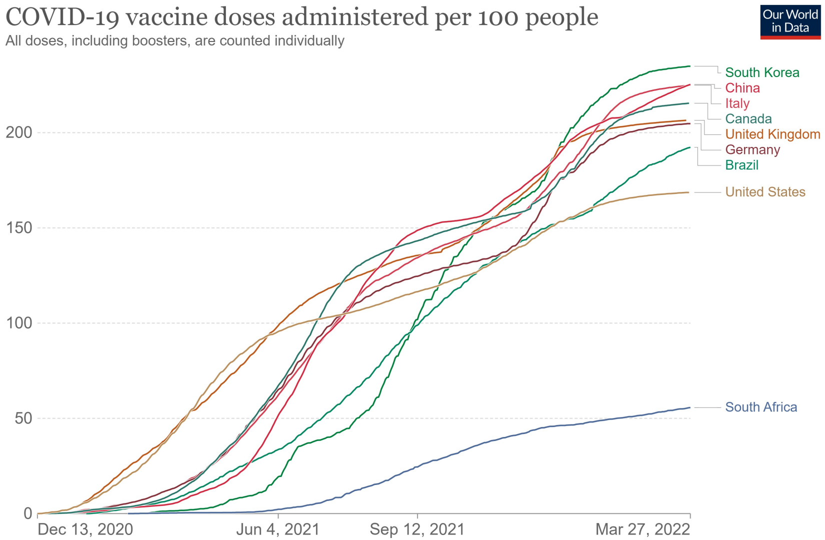


Source: Johns Hopkins University CSSE COVID-19 Data – Last updated 28 March, 19:05 (London time)
OurWorldInData.org/coronavirus • CC BY


Estimated cumulative excess deaths per 100,000 people during COVID-19
For countries that have not reported all-cause mortality data for a given week, an estimate is shown, with uncertainty interval. If reported data is available, that value only is shown. On the map, only the central estimate is shown.
Our World in Data
Central estimate
Upper bound, 95% uncertainty interval
Lower bound
China
United States
Brazil
Canada
United Kingdom
South Korea
Germany
South Africa
Italy
400
200
0
Jan 1, 2020
Mar 21, 2022
Source: The Economist (2022)
OurWorldInData.org/coronavirus • CC BY

COVID-19 vaccine doses administered per 100 people

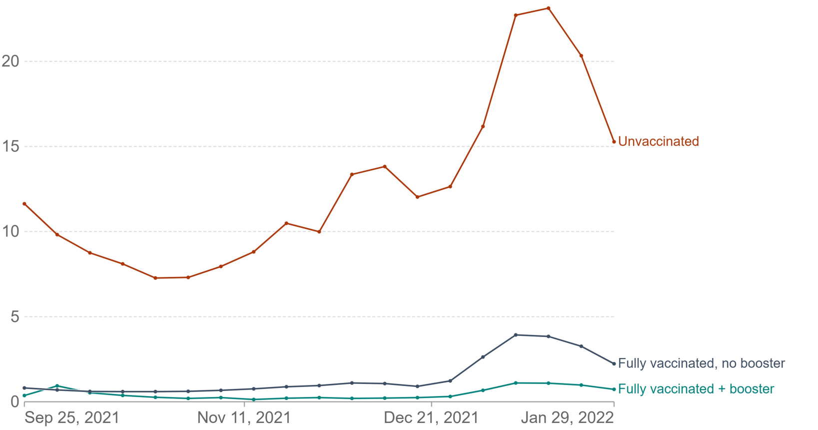
All doses, including boosters, are counted individually



Source: Official data collated by Our World in Data – Last updated 28 March 2022, 11:20 (London time) OurWorldInData.org/coronavirus • CC BY

United States: COVID-19 weekly death rate by vaccination status, All ages

Death rates are calculated as the number of deaths in each group, divided by the total number of people in this group. This is given per 100,000 people.



Source: CDC COVID-19 Response, Epidemiology Task Force

Note: Unvaccinated people have not received any dose. Partially-vaccinated people are excluded. Fully-vaccinated people have received all doses prescribed by the initial vaccination protocol. The mortality rate for the 'All ages' group is age-standardized to account for the different vaccination rates of older and younger people.

Supply or Demand Shock? Very hard to identify!

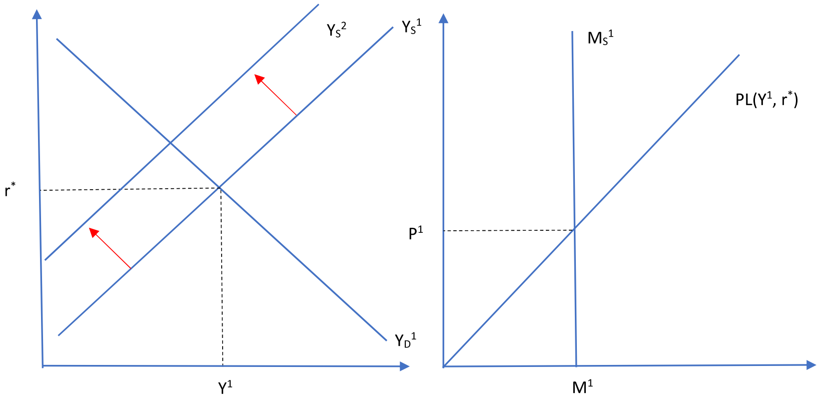
Why demand shock?

- People stop buying stuff, stop going out, stop travelling.
- Mainly affects services: restaurants, hotels, barbershops etc.
- Uncertainty about future dampens firm investment.

Why supply shock?

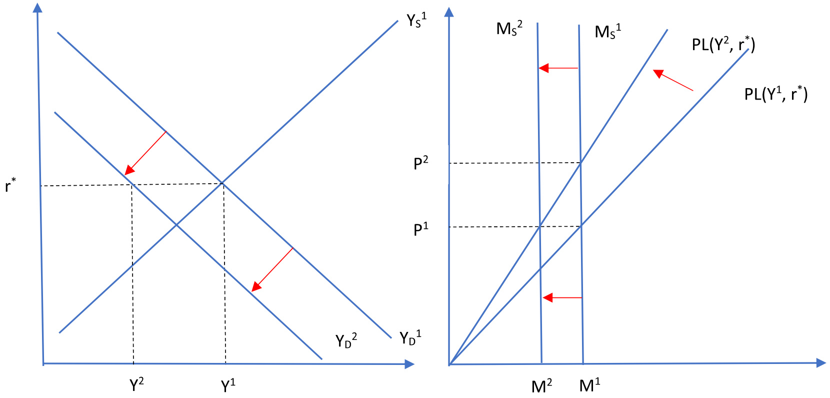
- People stopped working in many professions, negative labor supply shock.
- Distorted supply chains. Lack of parts for industries, broken global supply chains especially between Asia and Europe.

COVID-19 through the lens of the sticky price model



- Negative supply shocks shifts $Y_S^1 \rightarrow Y_S^2$.
- Output is demand driven here, still Y_1. Economy produces above it's capacity (negative output gap).

COVID-19 through the lens of the sticky price model

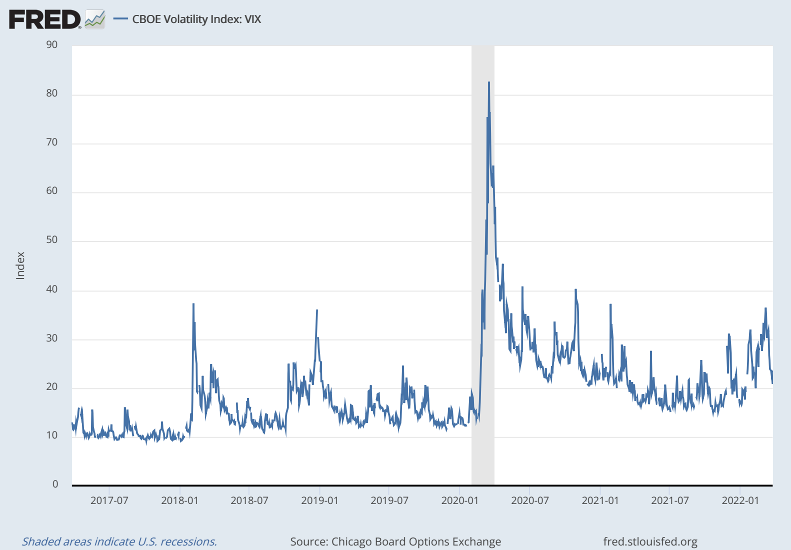


- Negative demand shocks shifts $Y_D^1 \to Y_D^2$.
- With unchanged target interest rate r^* money supply has to go down, given prices are fixed at P_1 in the short run.

So what do we see in the data so far?

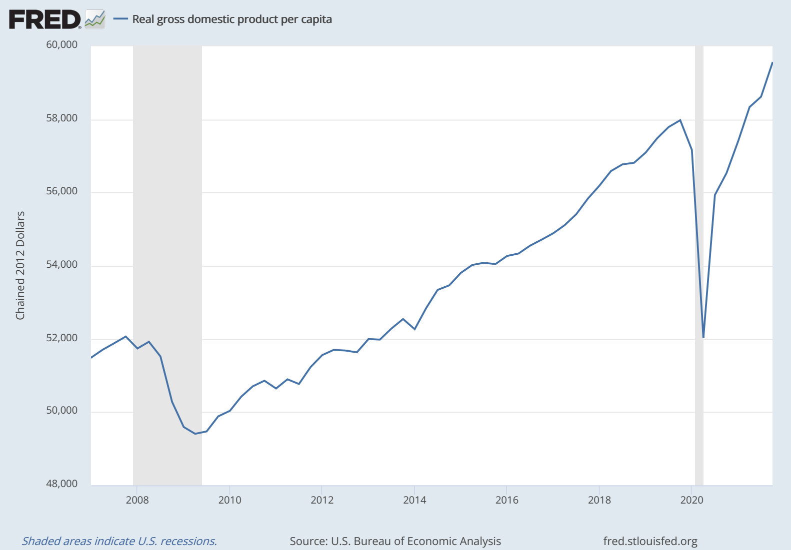
- One year into the COVID pandemic, already a lot has happened.
- Financial markets usually respond the fastest:
  - Stock market felt off the cliff but rebounded quickly.
  - Uncertainty about the prospects of the economy rose, but now uncertainty levels back to normal.
- Real economy:
  - Sharp drop in GDP and rapid increase unemployment. The effects seem to be rather short-lived though.
  - Very different picture from the Great Recession.

Uncertainty rose at the beginning of the pandemic.



Stock market tanked and wiped out all the gains since 2016 early in the pandemic.

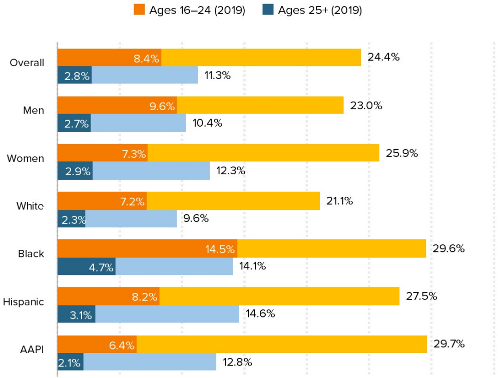
GDP also sharply declined, though very differently from the Great Recession



Unemployment Rate spiked but went back quickly.

Unemployment skyrocketed for young workers in the COVID-19 labor market

Unemployment rates in the spring of 2019 and 2020, by age, gender, and race/ethnicity



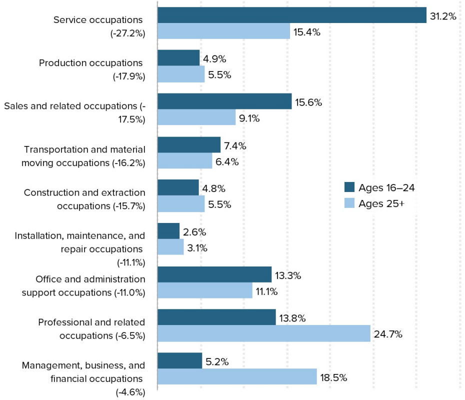
Notes: Unemployment rates are compared using a pooled average of April, May, and June data in each year. Racial and ethnic categories are mutually exclusive. Hispanic refers to Hispanic/Latinx of any race while white, Black, and AAPI refers to non-Hispanic whites, non-Hispanic Blacks, and non-Hispanic Asian Americans/Pacific Islanders, respectively.

Source: Economic Policy Institute Current Population Survey Extracts, Version 1.0.9 (2020), https://microdata.epi.org.

Economic Policy Institute

Young workers are heavily represented in the occupations most affected by COVID-19 shutdowns

Shares of workers ages 16–24 and ages 25+ in major occupations, 2019; occupations ranked by percent change in employment between February and May 2020

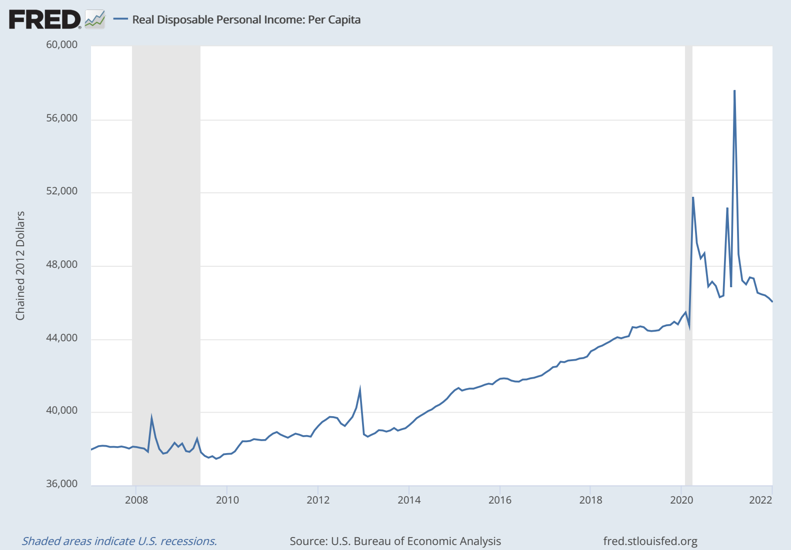


Note: The category farming, fishing, and forestry occupations is omitted because it accounts for less than 1% of total employment.

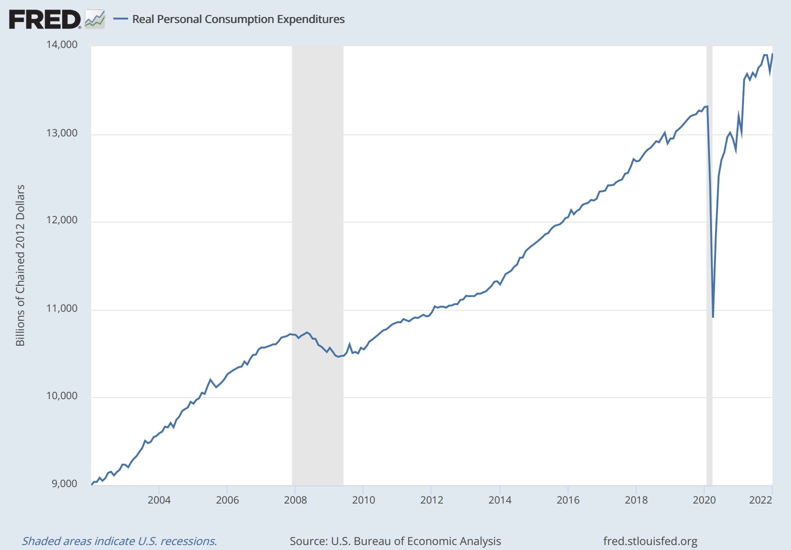
Source: Authors' analysis of Bureau of Labor Statistics Current Employment Statistics and Economic Policy Institute Current Population Survey Extracts, Version 1.0.9 (2020), https://microdata.epi.org.

Economic Policy Institute

Real Income improved, more than in the Great Recession.



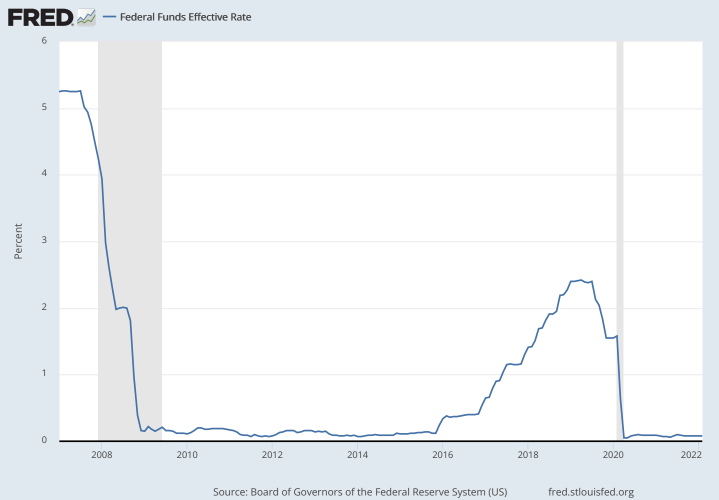
Consumption declined sharply, again different pattern compared to the Great Recession.



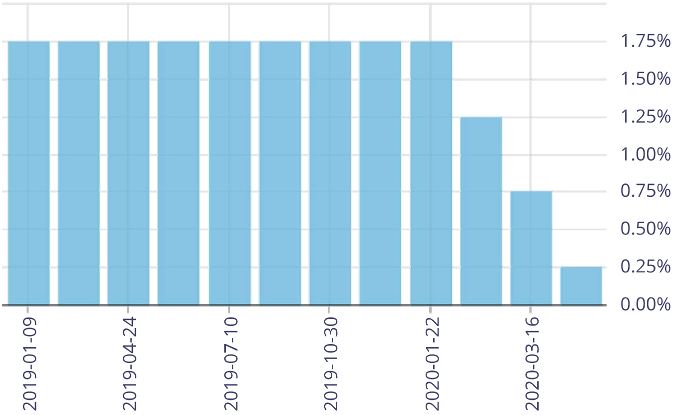
Macroeconomic Policy Response

1. Classical monetary policy response: interest rate cuts
2. Fiscal stimulus plans - helicopter money.
3. Financial market interventions.
4. Liquidity provisions and debt reliefs for households and firms.
5. Labor market policy tools.

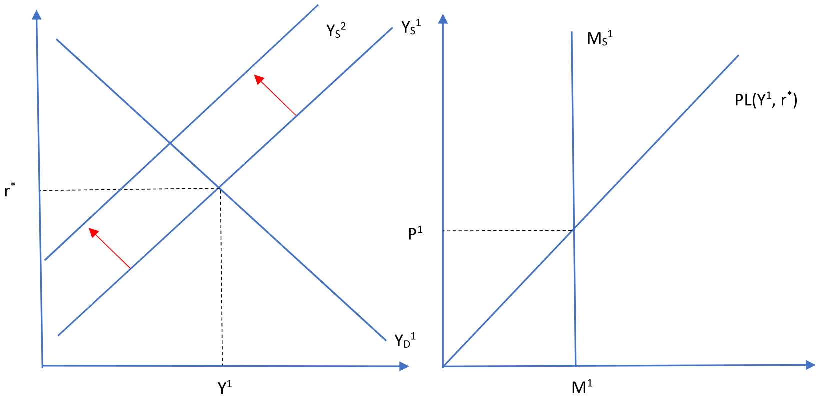
FED cut the target interest rate ...



... so did the Bank of Canada

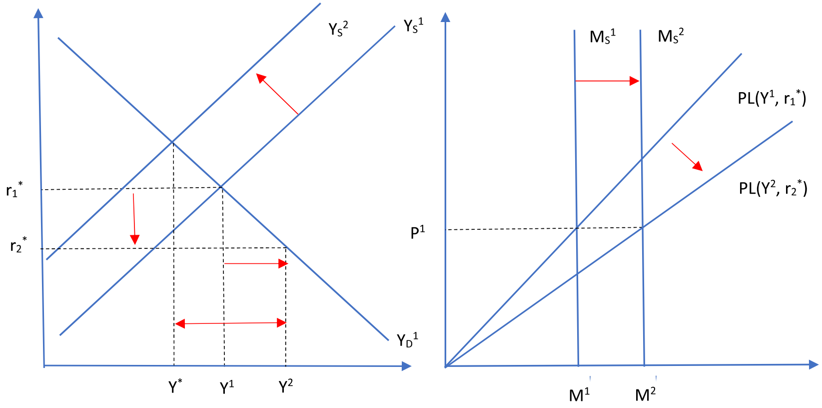


Monetary policy response to COVID-19 - supply shock



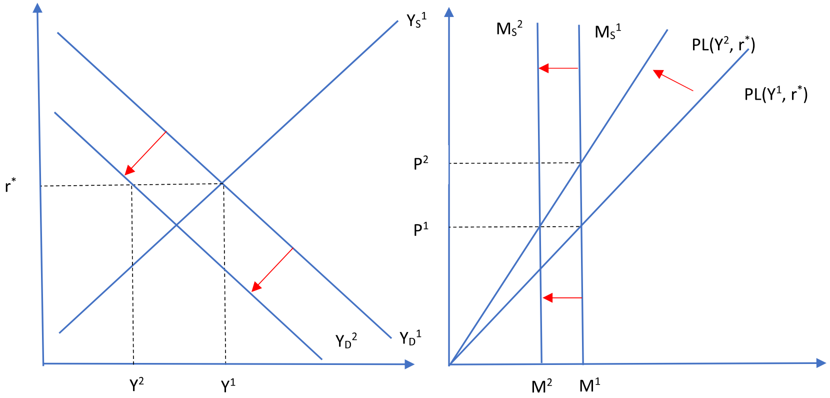
- Suppose first the shock is the negative supply shock.

Monetary policy response to COVID-19 - supply shock



- Reducing the target interest rate $r_2^* < r_1^*$.
- Negative output gap rises! $(Y^* - Y_2) < (Y^* - Y_1) < 0$. The economy is overheated with shortages of supply!
- In the long run as prices adjust, higher inflation and low output: **stagflation**.

Monetary policy response to COVID-19 - demand shock



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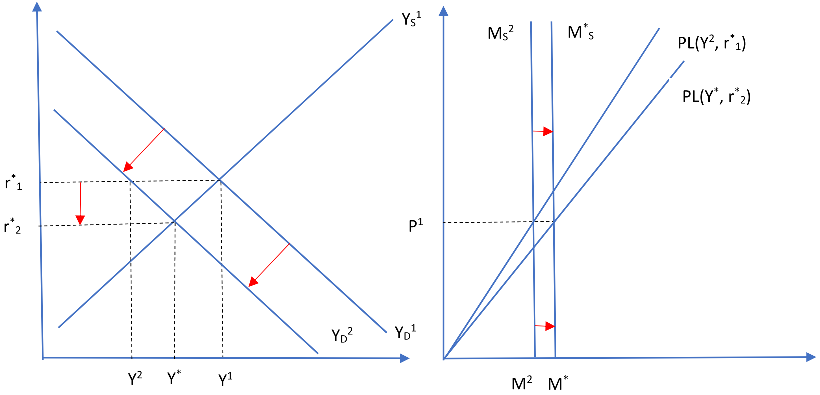
Monetary policy response to COVID-19 - demand shock



- Lowering the target rate to r_2^* restores efficiency but with lower output than the initial one, i.e. $Y^* < Y_1$.

The Coronavirus Aid, Relief, and Economic Security Act (CARES) - $2trn bazooka, one-tenth of GDP

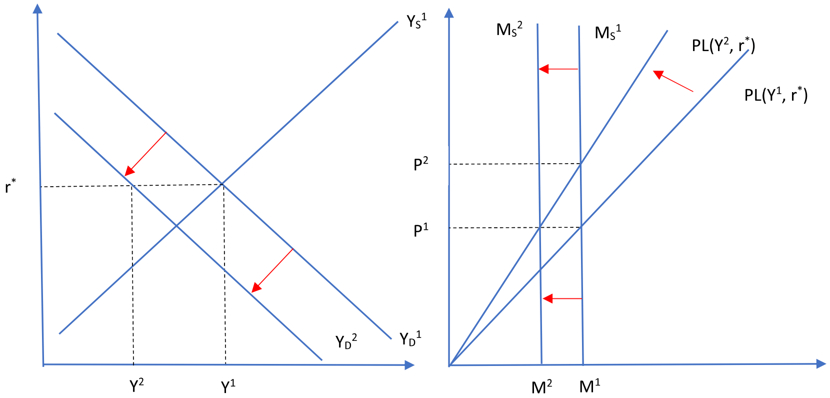
1. $250bn: sending cheques to Americans:
   - Below some generous income thresholds ($75,000 a year for a single person and $150,000 for a married couple) every family can expect $1,200 per adult and $500 per child.

2. $260bn: extending the safety net:
   - The federal government would pay to top up unemployment-benefit levels by $600 a week (current weekly average is $385).
   - Extended coverage: independent contractors, such as gig-economy workers. Those who have been laid off but not fired could receive compensation for lost hours.
   - Unemployment insurance benefit period would be extended from the usual 26 weeks to 39 weeks.

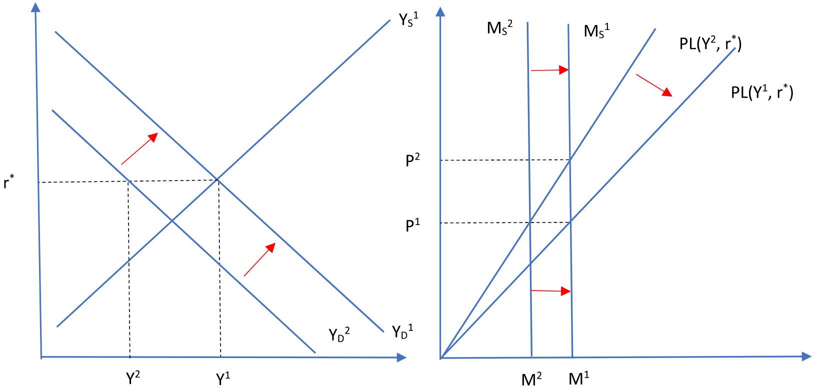
US Fiscal Stimulus - $2trn bazooka, one-tenth of GDP

3. $500bn: to stabilise firms and states through lending.
4. $350bn: set aside to save small and medium-size firms (those with fewer than 500 employees)
   - loans of up to $10m without interest or fees to pay for employees, rental costs and sundry other expenses.
5. $75bn: set aside to bail out embattled big firms like airlines and those deemed critical to national security.

Fiscal Stimulus following the demand shock

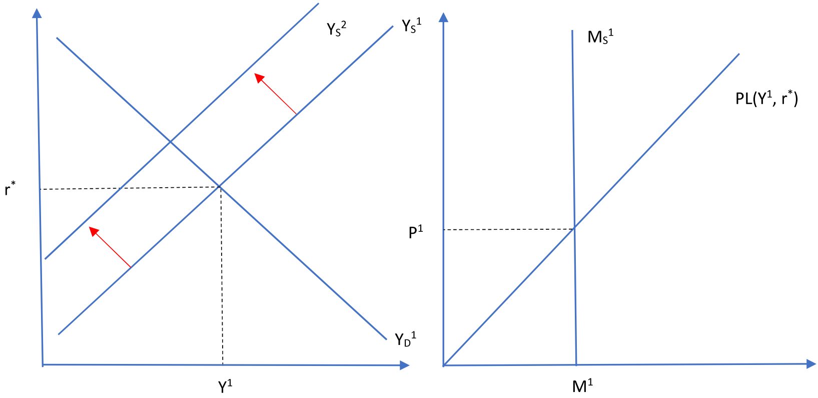


Fiscal Stimulus following the demand shock

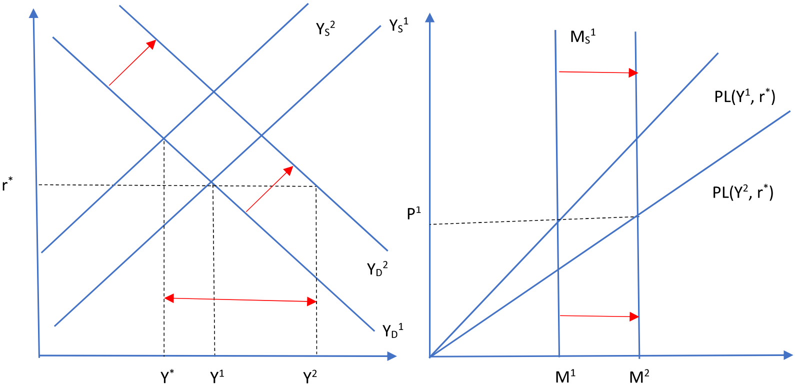


- Assume fixed labor supply upwards, i.e. intertemporal substitution of leisure is shut down (it's impossible to make people work harder with lockdown restrictions)
- Fiscal expansion can restore efficiency at the cost of crowding out private consumption - recall the logic of demand multiplier from Chapter 11.

Fiscal Stimulus following the supply shock

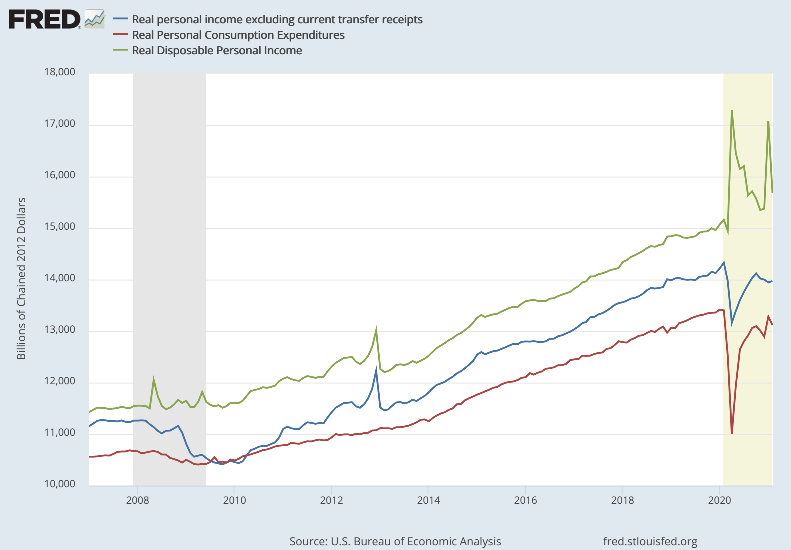


Fiscal Stimulus following the supply shock

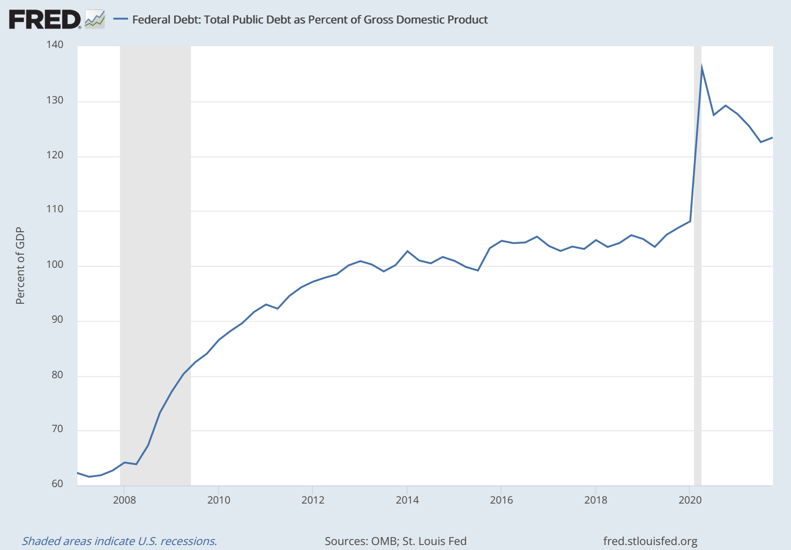


- Fiscal expansion induces more negative output gap, the new output is now Y_2. Economy is overheated, given shortages of the supply!
- Monetary policy has to accomodate here to keep the target rate unchanged. Both fiscal and monetary expansion in this case would make the problem worse and move the economy further away from efficiency.

Fiscal Stimulus increased transfers...



… at the cost of rising government debt!



Aggressive actions by the U.S. Federal Reserve (FED)

1. Near-Zero Interest Rates
2. Supporting Financial Market Functioning
3. Encouraging Banks to Lend
4. Supporting Corporations and Businesses
5. Supporting State and Municipal Borrowing
6. Cushioning U.S. Money Markets from International Pressures

Near-Zero Interest Rates

- **Federal funds rate:** The Fed has cut its target for the federal funds rate, the rate banks pay to borrow from each other overnight, by a total of 1.5 percentage points since March 3, bringing it down to a range of 0% to 0.25%.
- **Forward guidance:** the Fed has offered forward guidance on the future path of its key interest rate, saying that rates will remain low "until labor market conditions have reached levels consistent with the Committee's assessments of maximum employment and inflation has risen to 2 percent and is on track to moderately exceed 2 percent for some time."

Supporting Financial Market Functioning

- **Securities purchases (QE):** Treasury and mortgage-backed securities markets have become dysfunctional since the outbreak of COVID-19, and the Fed's actions aim to restore smooth market functioning so that credit can continue to flow. Between mid-March and early December, the Fed's portfolio of securities held outright grew from 3.9*trillionto*6.6 trillion.
- **Lending to securities firms:** Through the Primary Dealer Credit Facility (PDCF), a program revived from the global financial crisis, the Fed will offer low interest rate (currently 0.25%) loans up to 90 days to 24 large financial institutions known as primary dealers. The goal is to keep the credit markets functioning at a time of stress when institutions and individuals are inclined to avoid risky assets and hoard cash.
- **Backstopping money market mutual funds:** The Fed has re-launched the crisis-era Money Market Mutual Fund Liquidity Facility (MMLF), which lends to banks against collateral they purchase from prime money market funds, the ones that invest in corporate short-term IOUs known as commercial paper, as well as in Treasury securities.
- **Repo operations:** The Fed has vastly expanded the scope of its repurchase agreement (repo) operations to funnel cash to money markets and is now essentially offering an unlimited amount of money.

Encouraging Banks to Lend

- **Direct lending to banks:** The Fed lowered the rate that it charges banks for loans from its discount window by 2 percentage points, from 2.25% to 0.25%, lower than during the Great Recession. These loans are typically overnight—meaning that they are taken out at the end of one day and repaid the following morning—but the Fed has extended the terms to 90 days. The cash allows banks to keep functioning: depositors can continue to withdraw money, and the banks can make new loans.
- **Temporarily relaxing regulatory requirements:** The Fed is encouraging banks—both the largest banks and community banks—to dip into their regulatory capital and liquidity buffers, so they can increase lending during the downturn.

Supporting Corporations and Businesses

- **Direct lending to major corporate employers:** Fed on March 23 established two new facilities to support highly rated U.S. corporations. *The Primary Market Corporate Credit Facility (PMCCF)* allows the Fed to lend directly to corporations by buying new bond issuances and providing loans. Borrowers may defer interest and principal payments for at least the first six months so that they have cash to pay employees and suppliers. the new *Secondary Market Corporate Credit Facility (SMCCF)*, the Fed may purchase existing corporate bonds as well as exchange-traded funds investing in investment-grade corporate bonds.
- **Commercial Paper Funding Facility (CPFF):** Through the CPFF, another reinstated crisis-era program, the Fed buys commercial paper, essentially lending directly to corporations for up to three months at a rate between 1 to 2 percentage points higher than overnight lending rates.
- **Supporting loans to small- and mid-sized businesses:** The Fed's Main Street Lending Program, announced on April 9 and subsequently expanded and broadened to include more potential borrowers, aims to support businesses too large for the Small Business Administration's Paycheck Protection Program (PPP) and too small for the Fed's two corporate credit facilities. Fed will fund up to $600 billion in five-year loans. Businesses with up to 15,000 employees or up to $5 billion in annual revenue can participate.

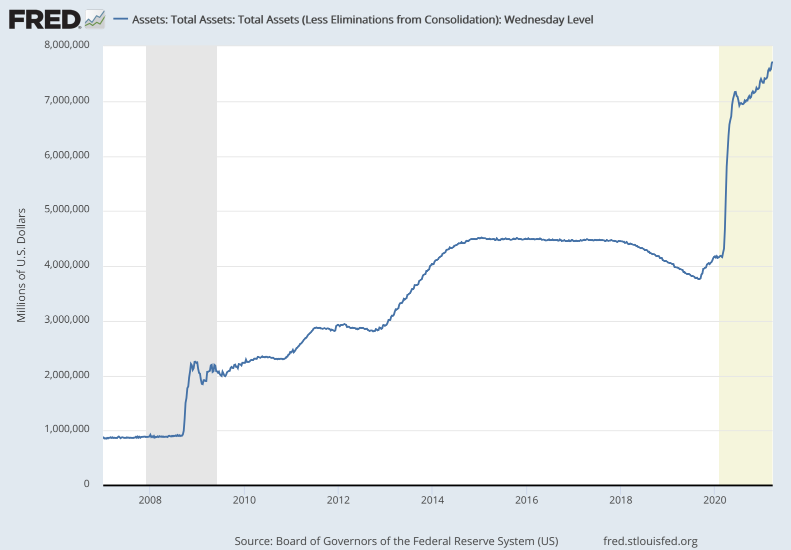
FED Loans Outstanding as of January 2021.

Federal Reserve 13(3) Facilities: Loans Outstanding

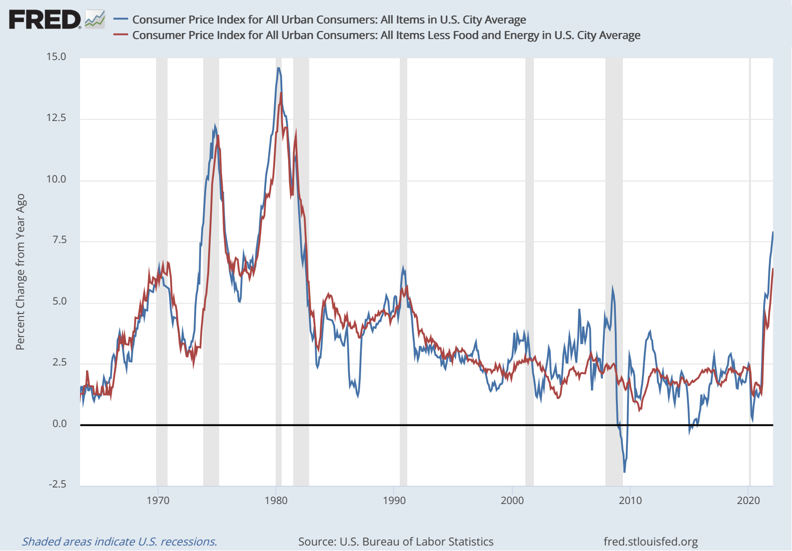
	Maximum Amount Authorized ($bns)	Total loans outstanding ($bns)	Updated
Commercial Paper Funding Facility	-	0.0	20-Jan
Main Street Lending Program	600	16.6	20-Jan
Money Market Mutual Fund Liquidity Facility	-	1.9	20-Jan
Municipal Liquidity Facility	500	6.3	20-Jan
Paycheck Protection Program Liquidity Facility	659	47.4	20-Jan
Primary Dealer Credit Facility	-	0.5	20-Jan
Corporate Credit Facility	750	14.1	20-Jan
Term Asset-Backed Securities Loan Facility	100	3.7	20-Jan

Note: This table will be updated monthly.

FED's Assets is skyrocketing...



...and inflation is rising



Conclusions

- We have been witnessing the worldwide recession and once in a century pandemic.
- Very aggressive economic policies have been implemented early on. The lesson from the Great Recession has been learnt. Only time will tell if they are appropriate.
- Surely more supply oriented policies should help.
- High inflation coming to the world economy.
- Thanks to your hard work during the course you are equipped now to critically review the policies!